

SALES PERFORMANCE & CUSTOMER BEHAVIOR ANALYSIS

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Business Problem

The company operates an e-commerce retail business in the United States, offering products across three main categories: Furniture, Office Supplies, and Technology. Despite having access to detailed transactional data, the company lacks a clear understanding of sales performance drivers, customer purchasing behavior, and product demand patterns over time. Additionally, limited visibility into seasonal trends, customer segment contributions, and shipping method performance restricts the company's ability to optimize marketing and operational strategies.

The management team tasked the data analyst with leveraging historical sales data to:

- Identify sales trends and seasonality
- Analyze customer behavior across segments
- Evaluate product and category performance
- Assess geographic and shipping-related patterns
- Detect anomalies or irregular sales behavior

The insights from this exploratory analysis will support data-driven marketing and business decisions, helping improve revenue performance and better align offerings with customer demand.

Key Business Questions

Sales Trends

- How have sales evolved over time?
- Are there identifiable seasonal patterns or fluctuations in sales?
- Which regions, states, and cities drive the highest and lowest sales?

Product Insights

- Which product categories and sub-categories generate the most and least sales?
- Which individual products are top-performing or underperforming?

Customer Insights

- How is the customer base distributed geographically?
- How do different customer segments contribute to total sales?
- Which customers generate the highest sales value?
- To what extent does repeat purchasing indicate customer loyalty?

Shipping Insights

- Which shipping methods are most frequently used and how do they contribute to sales?

Dataset Overview

The dataset includes the following variables:

- | | |
|---|---|
| • Row ID | • City |
| • Order ID | • State |
| • Order Date | • Postal Code |
| • Ship Date | • Region |
| • Ship Mode: Same Day, First Class, Second Class, Standard Class. | • Product ID |
| • Customer ID | • Category: Furniture, Office Supplies, Technology. |
| • Customer Name | • Sub-Category |
| • Segment: Consumer, Corporate, Home Office. | • Product Name |
| • Country: only the United States. | • Sales |

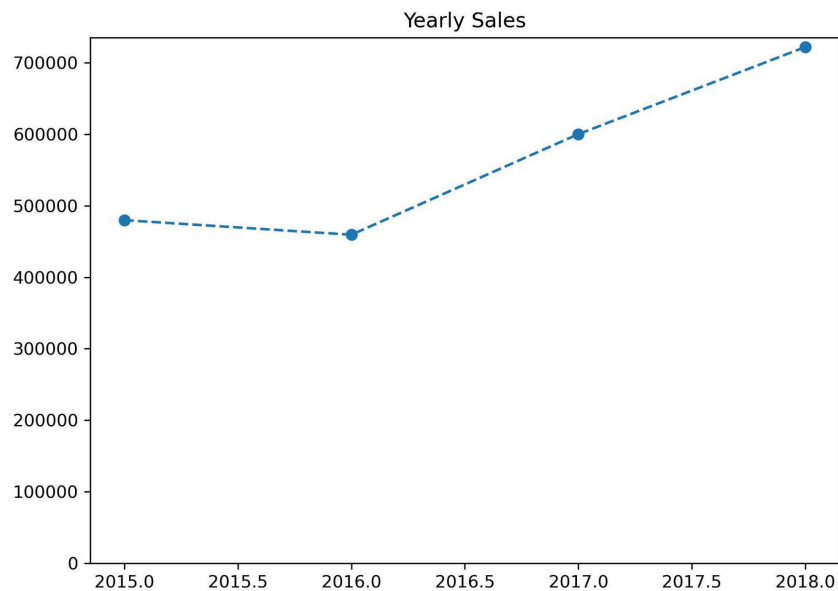
ANALYSIS AND FINDINGS

The dataset contains transactional data covering the years 2015 through 2018. It includes 4,922 sales distributed to clients across the United States.

SALES TRENDS

Sales performance was analyzed using both total revenue and order volume. Since the dataset contains multiple rows per order, order volume was calculated by counting unique Order IDs to avoid inflated results.

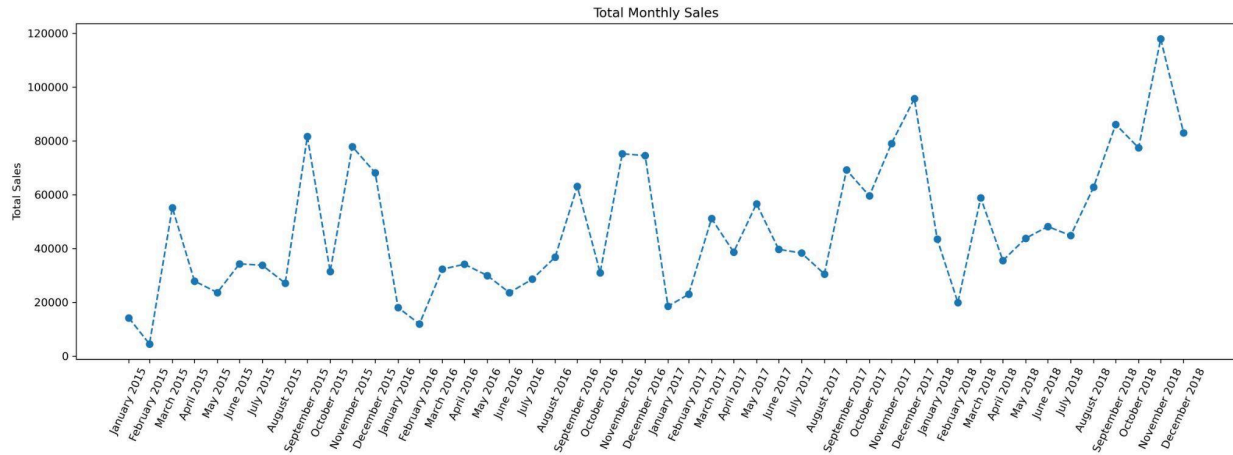
Sales revenue grew by 30% in 2017 compared to the relatively stable levels observed in 2015 and 2016, followed by an additional 20% increase in 2018. Overall, this represents a 50% total growth in sales compared to the baseline years of 2015–2016.



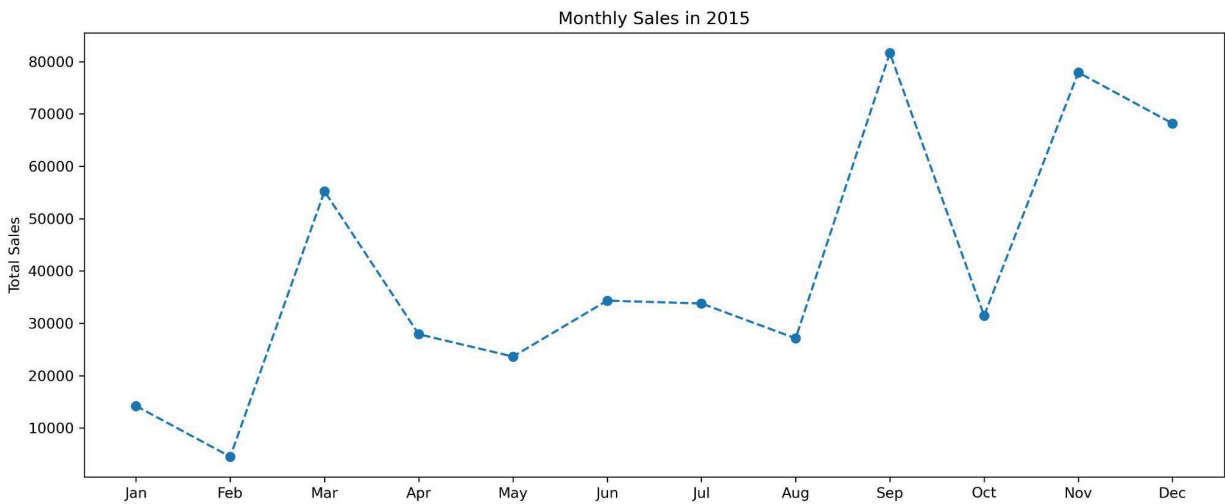
Sales revenue shows noticeable fluctuations throughout the year. However, a key finding is that the same seasonal pattern repeats consistently every year. January marks the slowest period, followed by a significant increase in March. From April to August, sales remain relatively stable, before rising sharply in September. Sales then decline slightly in October and peak again in November and December, before returning to their lowest levels in January.

Beyond this recurring seasonal pattern, there is a clear upward shift in sales revenue across all months over time, consistent with the growth observed at the annual level. This indicates that growth might be driven by a general increase in demand, customer base, or brand awareness, rather than by isolated high-performing months.

Without additional data on marketing campaigns, pricing strategies, or competitor activity, it is not possible to definitively explain the underlying drivers of this growth. It may represent sustained business expansion, a recovery from previous stagnation, or a combination of both. Further contextual data would be required to validate these hypotheses.



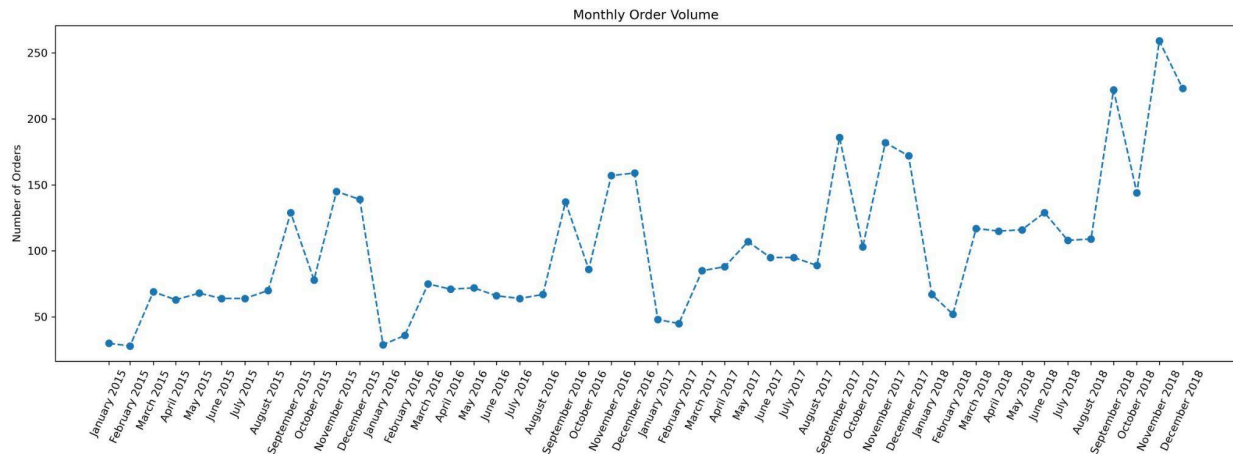
The following chart illustrates sales revenue performance for 2015, allowing the annual seasonal pattern to be observed more clearly.



Seasonality: Revenue vs Order Volume

An interesting observation is that, in terms of revenue, sales generated during the peak months (September–December) are only slightly higher than those generated during the slower and stable months (January–August). For example, in 2015, revenue during slower and stable months totaled USD 221K, compared to USD 259K during peak months. Similarly, in 2018, slower months generated USD 357K, while peak months reached USD 364K.

Order volume closely mirrors revenue patterns and provides additional clarity regarding both seasonality and overall growth. From 2015 to 2018, total order volume increased by approximately 75%, indicating that revenue growth is primarily driven by an increase in the number of customer purchases, rather than by price increases or inflationary effects.



On average:

- Slow months process approximately 30 orders
- Stable months process around 66 orders
- Peak months reach an average of 122 orders

Sales Metrics

Due to the highly skewed distribution of order values, the average order value was not considered a reliable standalone metric. Instead, the analysis focuses on median order value, order value distribution, and revenue concentration across high-value orders to better represent actual purchasing behavior.

Order value distribution is highly right-skewed. While the median order value is approximately \$152, only 10% of orders exceed \$1,158, and just 1% surpass \$4,200. This indicates that a small fraction of high-value orders drive a large share of revenue, making the average order value an unreliable sole metric.

Geographical Analysis of Sales

Revenue distribution and order volume distribution show very similar percentages across regions, states, and cities. Rankings are identical at the regional level and largely consistent at lower geographic levels, with the top three locations remaining unchanged. This indicates that revenue differences are primarily driven by order volume rather than variations in order value, suggesting relatively homogeneous purchasing behavior across locations.

Regional sales are relatively balanced, with a slight concentration in the West (31.4%) and East (29.6%), which together account for over 60% of total revenue. Central and South regions contribute 21.8% and 17.2%, respectively.

Sales at the state level are highly concentrated, led by California (19.7%) and New York (13.6%). Together, these two states generate over one-third of total revenue, indicating a strong geographic dependency on a small number of large markets.

Top states:

- California: 19.73%
- New York: 13.55%
- Texas: 7.45%
- Washington: 5.98%

- Pennsylvania: 5.14%

At the lower end of the distribution, several states individually contribute less than 0.1% of total sales, reflecting a long tail of low-volume markets with minimal revenue impact.

Lowest states:

- North Dakota: 0.04%
- West Virginia: 0.05%
- Maine: 0.06%
- South Dakota: 0.06%
- Wyoming: 0.07%

City-level sales are even more concentrated, with New York City alone accounting for 11.2% of total revenue. The top five cities collectively represent approximately one-third of overall sales, highlighting the importance of major urban centers.

Top cities

- New York City: 11.16%
- Los Angeles: 7.67%
- Seattle: 5.13%
- San Francisco: 4.82%
- Philadelphia: 4.81%

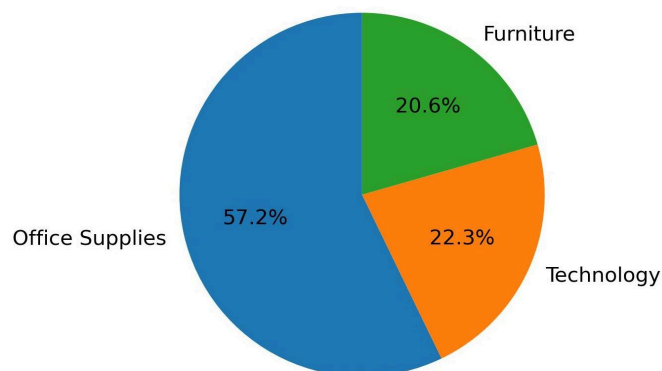
Conversely, a large number of cities contribute a negligible share of revenue individually, reinforcing the highly skewed nature of geographic sales distribution at the city level.

PRODUCT INSIGHTS

The dataset contains records for 1,849 distinct products that were sold during the analyzed period, classified into three main categories: Office Supplies, Furniture, and Technology.

It is important to note that this figure reflects only products with recorded sales transactions within the dataset timeframe. Since no inventory dataset is available, it is possible that additional products exist in the company's full catalog but were not sold during the analyzed period and therefore do not appear in this analysis.

Product Portfolio Share by Category



The Technology category has four subcategories:

- Phones
- Accessories
- Machines
- Copiers

The Furniture category also has four subcategories:

- Furnishings
- Chairs
- Tables
- Bookcases

The Office Supplies category has nine subcategories:

- Paper
- Binders
- Art
- Storage
- Appliances
- Labels
- Envelopes
- Supplies
- Fasteners

The dataset does not include product quantity or discount information. Additionally, the same product appears at varying price points across transactions, preventing the calculation of a standardized unit price.

Due to these constraints, product demand cannot be measured directly. Instead, product performance is evaluated using order frequency and revenue contribution as proxy indicators.

Order Frequency

It is observed that repeat purchases are infrequent across all products. Within a single year, many products were purchased only once. Given this information, and without access to inventory data, it can be inferred that some products may not be purchased at all during certain periods.

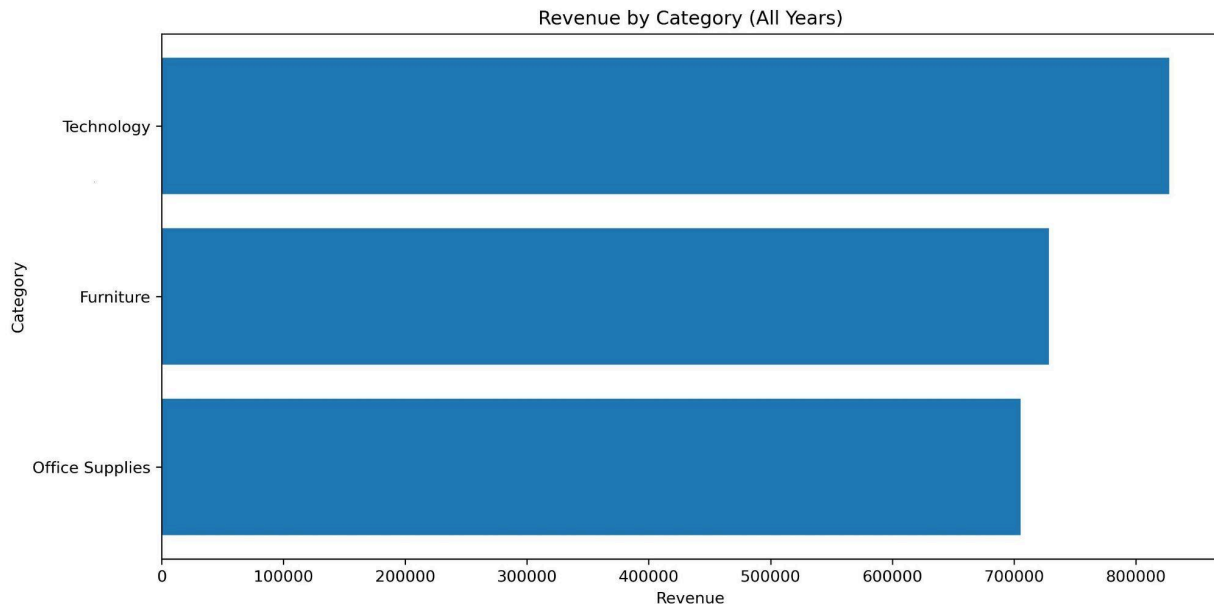
In the Furniture and Technology categories, the most frequently purchased products were sold only 6 to 7 times within a year. In contrast, within the Office Supplies category, lower-priced items such as staples and envelopes show higher repeat purchase frequency, reaching between 11 and 14 purchases annually, before sharply declining across the remaining product list.

Considering that the company serves clients nationwide, these figures suggest that many products are not even sold once per client per year. This may be concerning, particularly given the financial burden of maintaining a large and diversified inventory. The current sales frequency may not justify the cost of holding such an extensive product portfolio.

These findings could indicate either an overextended product assortment or inefficiencies in sales generation efforts. Further investigation into inventory strategy and sales performance may be warranted.

Revenue Contribution

At the Category level, Technology consistently generates the highest share of revenue, contributing approximately 36% of total sales. However, Furniture and Office Supplies follow closely, each accounting for roughly 32%. This distribution remains relatively stable across all analyzed years, with only minor fluctuations in category performance. Overall, revenue is fairly evenly distributed among the three main categories.

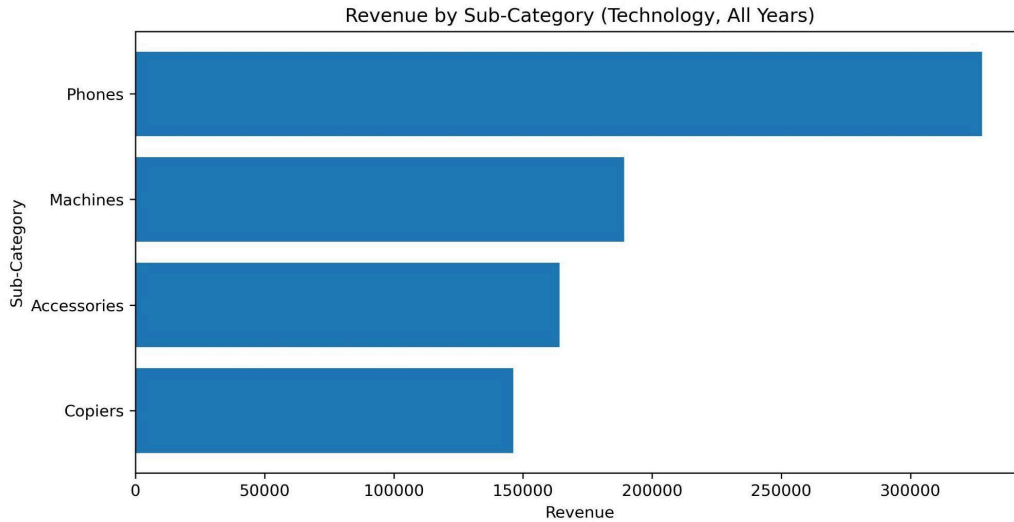


At the Sub-Category level, Technology is consistently led by Phones, which account for between 35% and 45% of category revenue annually.

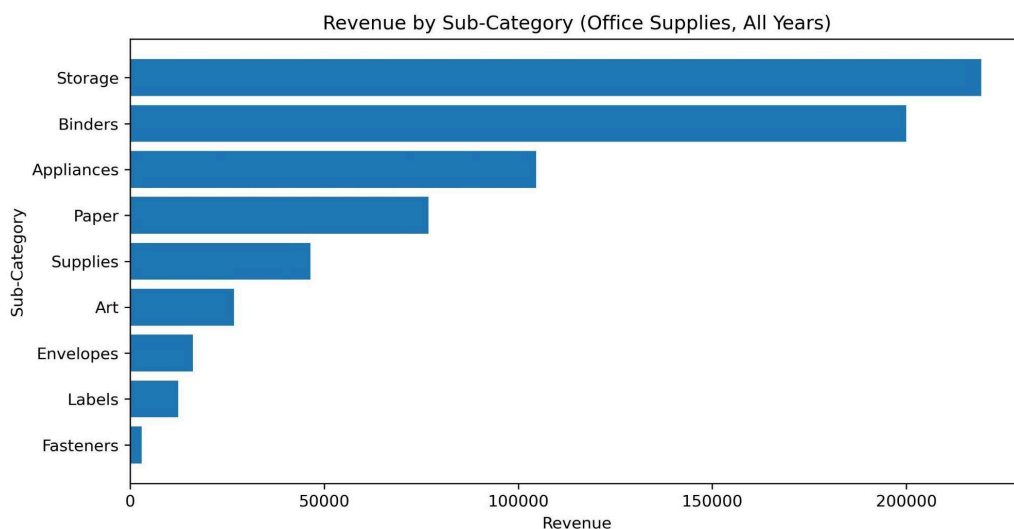
Machines ranks second in revenue contribution; however, its performance has been notably volatile. Its revenue share declined from 36% in 2015 to 17% in 2016, increased to 25% in 2017, and dropped again to 16% in 2018.

Meanwhile, Copiers have gained relevance over time, increasing from just 6% of category revenue in 2015 to approximately 20% in subsequent years.

Accessories demonstrate a more stable contribution, consistently accounting for between 15% and 25% of revenue, similar to the steady performance observed in Phones.

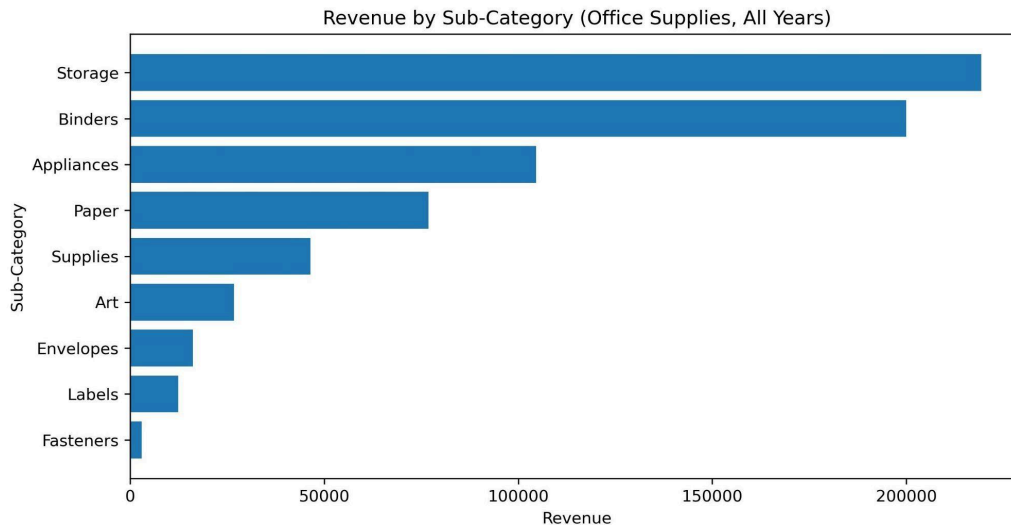


The Furniture category exhibits a stable and predictable revenue structure across its sub-categories. Chairs consistently lead, contributing approximately 40% to 50% of category revenue each year. They are followed by Tables at around 30%, Bookcases at 15% to 20%, and Furnishings at roughly 10%.



Office Supplies also displays a relatively stable revenue pattern. The leading sub-categories are Binders and Storage, each contributing around 30% of category revenue, followed by Appliances at approximately 15%.

The remaining sub-categories —Paper, Supplies, Art, Envelopes, Labels, and Fasteners— fluctuate in ranking year by year. However, Fasteners consistently generate the lowest revenue contribution across all years analyzed.



The presence of dominant sub-categories within each main category suggests that revenue performance is concentrated in specific product groups, which may represent key commercial drivers for the business.

CUSTOMER INSIGHTS

There is no customer dataset available, therefore, all customer insights are derived from sales records.

The company serves 793 unique customers across 49 states, indicating a nearly nationwide customer base. When analyzing customer count over the years, there has been an approximate 20% increase, coinciding with growth in both revenue and orders.

When analyzing unique customers across regions, states, and cities, the sum of customers at each geographic level exceeds the global customer count. This indicates that some customers place orders from multiple locations or use different shipping addresses. In fact, 779 customers purchased from more than one city, state, or region, demonstrating significant geographic overlap.

At the regional level, customer participation remains relatively stable year over year. On average:

- Approximately 35%–45% of customers placed at least one order in the Central region.
- Approximately 40%–50% placed at least one order in the East.
- Approximately 25%–35% placed at least one order in the South.
- Approximately 45%–55% placed at least one order in the West.

When analyzing all years combined, the data reveals substantial cross-region purchasing behavior. The broader time horizon increases geographic overlap, suggesting cumulative multi-region activity over time.

This is the number of unique customers per region during all years:

Region	Number of Customers	Share of Total Unique Customers (%)
Central	626	78.94
East	669	84.36
South	509	64.19
West	681	85.88

The states generating the highest customer activity are:

- California
- New York
- Texas
- Pennsylvania
- Illinois
- Washington
- Ohio
- Florida

The cities with the highest customer activity are:

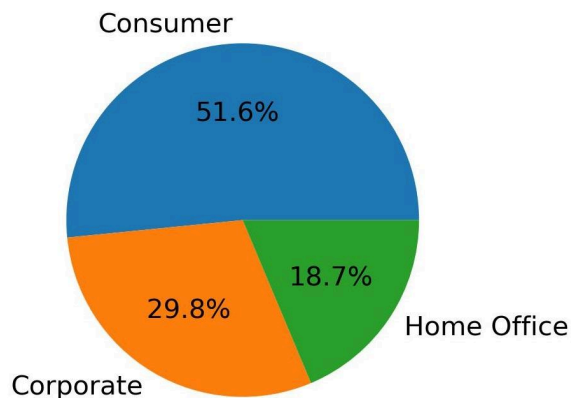
- New York City
- Los Angeles
- Philadelphia
- San Francisco
- Seattle
- Houston
- Chicago
- Columbus
- San Diego
- Dallas

Customer Segmentation

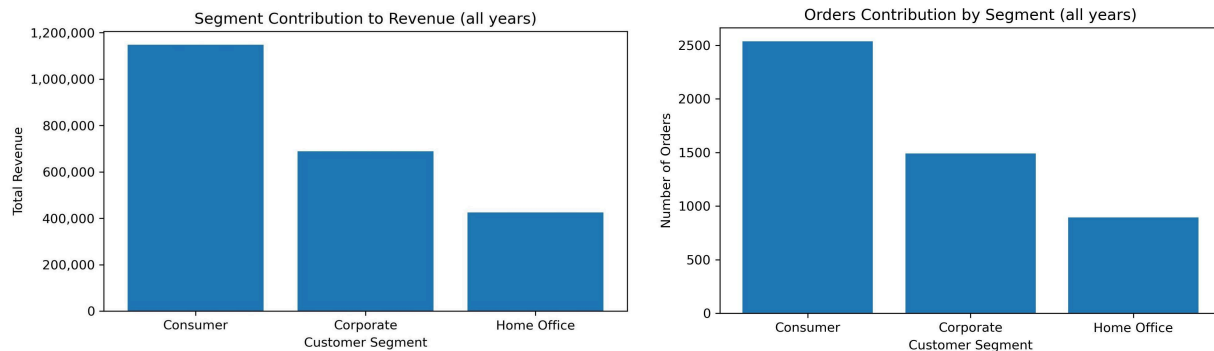
Customers are classified into three segments: Consumer, Corporate, and Home Office.

The customer base consists of approximately 50% Consumer clients, 30% Corporate clients, and 20% Home Office clients.

Customer Distribution by Segment (all years)



This distribution remains consistent across all years included in the dataset and across regions, indicating that it represents a stable structural characteristic of the customer base rather than a temporary fluctuation.



Customer, order, and revenue distributions are highly aligned across segments, indicating that performance differences might be driven by segment size rather than behavioral or spending differences.

Customer Loyalty

Repeat purchasing reflects transactional loyalty. While it demonstrates ongoing engagement, it does not necessarily reflect emotional attachment or long-term brand commitment. In this analysis, loyalty is assessed through purchase frequency and revenue contribution, providing an economic perspective on retention strength.

The analysis shows the presence of repeat buyers; however, no extreme concentration of purchasing behavior was observed. The highest-frequency customers placed between 6 and 9 orders per year.

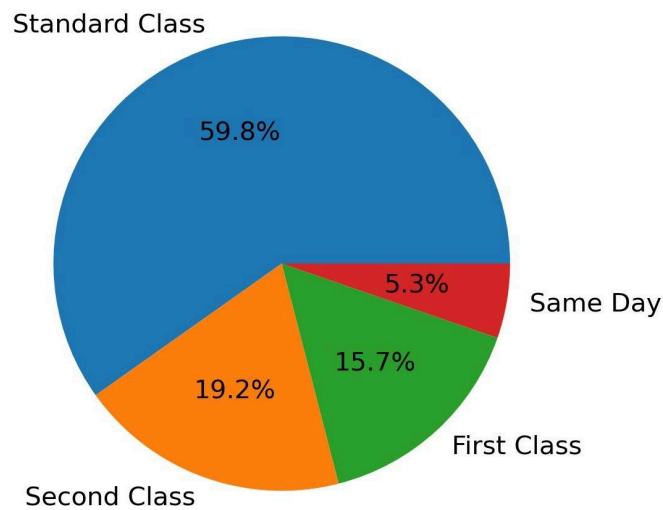
Overall, the findings suggest a stable and moderately retention-driven customer base, where repeat purchasing contributes to performance without creating strong revenue dependence on a small group of highly loyal customers.

SHIPPING INSIGHTS

The company operates with four shipping methods: Same Day, First Class, Second Class, and Standard Class.

Across the entire time period covered by the dataset, the share distribution of shipping methods remains very consistent year after year and across all states and regions (North, South, East, and West), showing a clear pattern in shipping preferences. Standard Class is the most frequently used method, accounting for around 60% of deliveries, followed by Second Class at approximately 20%, First Class at 15%, and lastly Same Day with only 5%. This indicates that faster or premium shipping services are not highly preferred by customers.

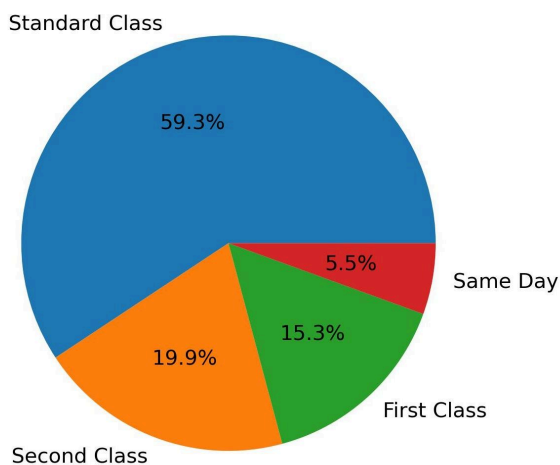
Shipping Methods Share Across Dataset



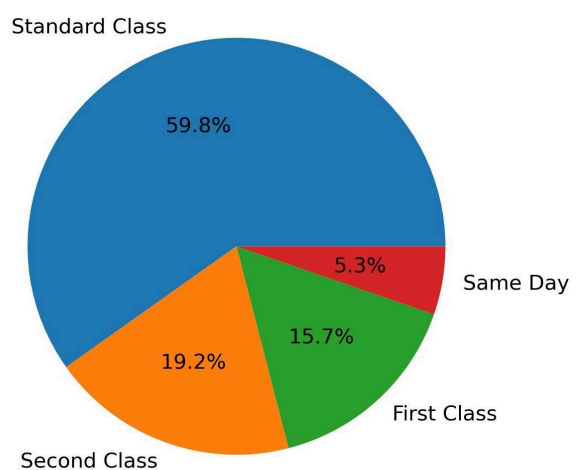
Without information regarding pricing or service quality differences among shipping methods, the underlying reasons behind the significant usage gap cannot be fully assessed.

The shipping method share distribution observed above closely mirrors the distribution of both total sales and order volume by shipping method. This alignment suggests that average order values are relatively consistent across shipping options, indicating that customers do not significantly adjust purchase value based on shipping speed.

Total Sales Share by Shipping Method



Total Orders Share by Shipping Method



Shipping Efficiency

The dataset does not include a data dictionary; therefore, the exact meaning of some variables cannot be definitively confirmed. It is inferred that the dataset contains Order Date (purchase date) and Ship Date (date when the order was dispatched), but does not include the package arrival date. Without an arrival date, shipping efficiency in terms of delivery timing cannot be assessed. If Ship Date were to represent the arrival date instead, the analytical scenario would change.

Shipping lead times increase progressively by service level, suggesting that Ship Mode reflects fulfillment priority. Standard Class orders take an average of 5 days before dispatch, with some taking up to 7 days, while Same Day orders are shipped almost immediately. First Class and Second Class orders take an average of 2 and 3 days, respectively. These patterns are consistent across all four regions and states.

DATA LIMITATIONS

The dataset does not include a column indicating the quantity of units purchased per transaction. As a result, multiple entries for the same product name may reflect different total sales values, and the unit price cannot be determined. Additionally, there is no information regarding discounts or promotions applied to individual sales. Consequently, unit pricing, price consistency, price segmentation, and true product demand cannot be accurately assessed, limiting any analysis involving pricing variables.

No cost data is available; therefore, profit-related metrics such as margin analysis and profitability by segment cannot be evaluated.

The dataset does not include a delivery or package arrival date. Without this information, delivery time cannot be measured, and shipping efficiency over time cannot be assessed to determine potential areas for operational improvement.

There is also no detailed information regarding the differences in service levels or pricing structures across shipping methods. In the absence of a data dictionary, interpretations of shipping categories remain limited and cannot be considered conclusive.

Finally, there is no separate customer dataset available. As a result, all customer insights are derived solely from transactional sales records.

MARKETING & BUSINESS RECOMMENDATIONS

SALES TRENDS

Further analyze the drivers behind sales growth from an expanding customer base — specifically whether it stems from an external increase in demand, marketing campaigns, pricing strategies, or competitor activity. This will help establish a baseline for planning marketing campaigns aimed at intentional business growth.

Identify the reasons behind seasonal sales trends. Understanding these patterns will also provide a foundation for planning targeted marketing campaigns and driving intentional business growth.

PRODUCT INSIGHTS

Excess inventory and very low repeat purchase rates of products are a concern. Inventory levels must be evaluated, as this is a key budget driver that may be negatively impacting the business.

To address these issues, a few targeted actions are recommended. First, promotional campaigns — such as bundle deals, limited-time discounts, or clearance initiatives — should be considered to move stagnant or slow-moving stock and reduce excess inventory carrying costs. Second, a customer loyalty or repurchase incentive program, such as reward points, subscription models, or exclusive returning-customer offers, could help stimulate repeat purchases and increase customer lifetime value. Finally, a thorough review of the product assortment is advisable to identify underperforming Stock Keeping Units (SKUs) that may be tying up capital without generating sufficient return, and to better align the offering with actual customer demand patterns observed in the data.

CUSTOMER INSIGHTS

Given that customers are distributed across the country and span cities of varying sizes, it would be worthwhile to segment customer service by assigning each sales representative a dedicated region and customer type (Corporate, Home Office, or Consumer). This approach would allow the team to tailor marketing efforts and product offerings to the specific needs and purchasing behaviors of each customer segment, while fostering loyalty through close, dedicated business relationships.

SHIPPING INSIGHTS

It is recommended that all packages be assigned the same priority level for dispatching. This would improve operational efficiency, strengthen brand perception and customer trust, and ultimately enhance the company's overall performance.

DATA LIMITATIONS

The sales transaction dataset should be expanded to include the following columns: units purchased per product, discounts and promotions applied, cost of goods sold (COGS), and actual package delivery date.

Additionally, to enable more robust future analysis, it would be valuable to collect a description of each shipping method and how it operates, as well as a comprehensive customer dataset including key information such as acquisition date, location, and other relevant attributes.